MATH 345 Skeleton Notes

Unit 5: Optimization and training

Section 5.1: Optimization problems and critical points

Optimization problems

In this unit, we apply multivariate calculus to solve some *optimization problems*.

Definition: Optimization problems Let $f : \mathbb{R}^n \to \mathbb{R}$ be a scalar-valued function, and consider a set $U \subseteq \mathbb{R}^n$. Then we can consider two optimization problems, in which we call f the

objective function of the problem, and U the

feasible region of the optimization problem:

- The minimization problem of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *smallest* among all elements of U (so the function is *minimized* on U by $\mathbf{x}^*$).
- The maximization problem of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *largest* among all elements of U (so the function is *maximized* on U by $\mathbf{x}^*$).

Example A common optimization problem that we have already encountered in The orthogonal projection onto a subspace involves taking a subspace $U \subseteq \mathbb{R}^n$ and a given vector $\mathbf{x}_0 \in \mathbb{R}^n$, and finding the vector $\mathbf{x}^* \in U$ that is *closest* to $\mathbf{x}_0$, i.e. finding the vector that minimizes the scalar-valued function $g : \mathbb{R}^n \to \mathbb{R}$ on U, defined by setting $g(\mathbf{x}) = \|\mathbf{x}_0 - \mathbf{x}\|$. Then $\mathbf{x}^*$ is the orthogonal projection of $\mathbf{x}_0$ onto U, and we have seen in Projection theorem that if U has an orthogonal basis $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$, then

$$\mathbf{x}^* = \frac{\mathbf{x}_0 \cdot \mathbf{f}_1}{\|\mathbf{f}_1\|^2}\mathbf{f}_1 + \frac{\mathbf{x}_0 \cdot \mathbf{f}_2}{\|\mathbf{f}_2\|^2}\mathbf{f}_2 + \cdots + \frac{\mathbf{x}_0 \cdot \mathbf{f}_m}{\|\mathbf{f}_m\|^2}\mathbf{f}_m.$$

In the section on Least squares from gradients, we return to least squares and derive the normal equations by setting a gradient equal to zero.

Note: Unit 4 connection In Unit 4, projection was a closest-point problem. If W is a subspace and $\mathbf{b}$ is a vector, the projection problem asks for the point $\hat{\mathbf{b}} \in W$ that minimizes

$$\|\mathbf{b} - \mathbf{w}\|$$

over all $\mathbf{w} \in W$. Unit 4 solved this using orthogonality. Unit 5 studies the same kind of question with gradients: minimize a function by looking at how it changes.

Activity: Projection as optimization

Let

$$W = \operatorname{span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right\} \subseteq \mathbb{R}^2, \qquad \mathbf{b} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

A general point of W has the form

$$\mathbf{w}(t) = \begin{bmatrix} t \\ 0 \end{bmatrix}.$$

1. Write the squared distance $g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2$.

2. Minimize $g(t)$.
3. What point of W is closest to $\mathbf{b}$?
4. How does this agree with the projection picture from Unit 4?

Tags. `[U5-LO1, U4-LO4 | C+R | Core]`

We have

$$\mathbf{b} - \mathbf{w}(t) = \begin{bmatrix} 2-t \\ 3 \end{bmatrix},$$

so

$$g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2 = (2-t)^2 + 9.$$

This is minimized when $t = 2$. Therefore the closest point in W is

$$\hat{\mathbf{b}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

This agrees with the Unit 4 projection picture: the residual

$$\mathbf{b} - \hat{\mathbf{b}} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

is orthogonal to W.

Critical points

Definition: Minima and maxima Suppose f is a scalar-valued function. A point $\mathbf{a}$ in the domain of f is a

- local minimum if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \leq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball, and also in the domain of f. In this case, we say $f(\mathbf{a})$ is a local minimum value for the function f.
- local maximum if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \geq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball, and also in the domain of f. In this case, we say $f(\mathbf{a})$ is a local maximum value for the function f.
- global minimum if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f. In this case, $f(\mathbf{a})$ is the global minimum value, or absolute minimum value, of the function f.
- global maximum if $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f. In this case, $f(\mathbf{a})$ is the

$\boxed{\text{global maximum value}}$, or

$\boxed{\text{absolute maximum value}}$, of the function f.

A local minimum or local maximum is also called a

$\boxed{\text{local extremum}}$.

Remark The graph of the function $-f$ is obtained by reflecting the graph of the function f in the output axis. This operation turns local and global maxima into local and global minima, and vice versa. So methods for finding minima for a general function often give methods for finding maxima, which makes the two problems often equivalent.

Definition: Critical points Suppose f is a scalar-valued function that is differentiable at a point $\mathbf{a}$. Then we say $\mathbf{a}$ is a

$\boxed{\text{critical point}}$ if $\nabla f(\mathbf{a}) = \mathbf{0}$.

For a scalar-valued function $f(x, y)$, a critical point is a solution to the two equations $f_x(x, y) = 0$ and $f_y(x, y) = 0$.
Just as the derivative of a scalar-valued function of a single variable helps identify local minima and maxima, partial derivatives provide information about possible local minima/maxima.

Theorem Suppose f is a scalar-valued function that is differentiable at a point $\mathbf{a}$. If $\mathbf{a}$ is also local extremum of f, then $\mathbf{a}$ is a critical point of f.

Activity

Find the critical points of the function

$$f(x, y) = x^2 + y^2 - 2x + 4y.$$

To find the critical points, we calculate the partial derivatives and set them equal to zero. We find that

$$f_x(x,y) = 2x - 2 \quad \text{and} \quad f_y(x,y) = 2y + 4.$$

The critical points are the pairs (x, y) that simultaneously solve the equations $2x - 2 = 0$ and $2y + 4 = 0$. So the only critical point is $(1, -2)$.

Activity

Find the critical points of $f(x, y) = x^2 - y^2$.

We compute that $f_x(x, y) = 2x$ and $f_y(x, y) = -2y$. The only (x, y) such that $f_x(x, y) = 2x = 0$ and $f_y(x, y) = -2y = 0$ is $(0, 0)$. This is the only critical point.